SRIKUMAR VIKASHAN

Computer Engineering Undergraduate
University of Peradeniya

Profile

Computer Engineering undergraduate at the University of Peradeniya with hands-on experience in computer vision, deep learning, optimization, software development, and low-level systems programming. Built projects spanning CNN-based image restoration, sparse 3D reconstruction, deep-learning-based defect detection, scheduling optimization, and software 3D rendering. Interested in internship opportunities in **AI, Computer Vision, Machine Learning, and Software Engineering**.

Education

BSc Engineering (Honours) in Computer Engineering **2024 – 2028**
Faculty of Engineering, University of Peradeniya *Sri Lanka*

Relevant Coursework: Computer Architecture, Data Structures and Algorithms, Neural Networks and Deep Learning, Digital Image Processing, Database Systems, Computer Networks, Multivariable Calculus

Experience

Casual Instructor – Data Structures and Algorithms I **2026**
University of Peradeniya

- Supported undergraduate teaching activities for Data Structures and Algorithms I and assisted students with programming and fundamental algorithmic concepts.

Technical Skills

Languages:	Python, C/C++, Java, Dart, SQL, Verilog, ARM32 Assembly
AI / Computer Vision:	PyTorch, TensorFlow, OpenCV, scikit-learn, NumPy
Software Development:	Flutter, FastAPI, Flask, PostgreSQL
Tools:	Git, GitHub
Hardware / Networking:	Arduino, Digital Design, Cisco Switch Configuration, VLANs and Trunking

Selected Projects

Low-Light Image Denoising – MoraSPCup 2026 *PyTorch, Deep Learning*

- Developed a residual CNN in PyTorch for removing noise from low-light images and improving visual quality.
- Achieved **28.00 dB PSNR** and **0.809 SSIM** on the validation set.
- Implemented the model as part of a team submission for the MoraSPCup 2026 image-denoising competition.

Sparse 3D Reconstruction using Structure from Motion *Python, Computer Vision*

- Reconstructed sparse 3D scene geometry from multiple 2D photographs using an incremental Structure-from-Motion pipeline.
- Applied feature matching and camera-pose estimation to recover spatial relationships between images and reconstruct the scene in 3D.

Fabric Defect Detection and Localisation *Deep Learning, Computer Vision*

- Trained a deep neural network to identify defects in fabric images and localise the regions in which defects occur.
- Applied deep-learning-based image analysis techniques for automated visual quality inspection.

OptiFlow – Job Shop Scheduling System *Flutter, FastAPI, OR-Tools*

- Developed a scheduling application for print-shop jobs using Flutter for the frontend and FastAPI for backend services.
- Integrated Google OR-Tools to automatically generate schedules based on jobs and available resources.

libtiny3d – Software 3D Renderer *C*

- Built a software 3D graphics engine from scratch in C without using OpenGL or DirectX.
- Implemented transformations, lighting and animation to render 3D objects frame by frame, demonstrated using a rotating shaded wireframe soccer ball.

Achievements & Certifications

- **AgenTrix 2026** – Participated in a 12-hour hackathon and developed *CitiReady LK*, a multi-agent readiness auditor for government services.
- **Introduction to Cybersecurity** – Cisco Networking Academy, September 2026.
- **Databricks Accredited AI Agent Fundamentals** – Databricks, May 2026.

Languages

Tamil | English | Sinhala